

Regional model validation record

Field	Entry
Region	Northwest African Shelf / Canary Current, LME_027; Sea Around Us target boundary. Regional workbook
Catch source	Sea Around Us catch series 1950–2019. Reference: 2019 landings; all 512 exact labels, including unidentified and zero-catch records. Catch inputs Regional catch
Selected article	Morissette, Melgo, Kaschner, Gerber & Bamy (2009). Food web model and data for studying the interactions between marine mammals and fisheries in the Northwest African ecosystem. Fisheries Centre Research Reports 17(2). Exact final report unavailable; retained 2008 preliminary source is distinct. 2008 preliminary PDF Source identity review
Other known articles	Guénette et al. (2014), Mauritanian Shelf/Banc d’Arguin, DOI 10.1371/journal.pone.0094742: distinct 1991–2006 model family. Villanueva et al. (2004), An Ecopath model of the Sine-Saloum Delta biosphere reserve (Senegal): distinct 38-group estuarine model. 2014 article Retained alternatives
Selected model	27_118_Northwest_Africa_(1987); EcoBase accession 118, nominal 1987, 27 ecological groups plus synthetic diet_import. Broad Northwest African offshore ecosystem; reported area 3,561,029 km ² . Selected JSON Model folder
Other models from the same article	No additional temporal model from this exact final report is verified. The stored _1883ddd3 candidate is a separately archived identity, not a second source model. Banc d’Arguin/Mauritanian Shelf 1991 belongs to the separate Guénette source. Identity inventory
Selection rationale	Article: ? (no separate article-choice rationale recorded). Model: user selected this available non-failing 1987 alternative over Banc d’Arguin/Mauritanian Shelf 1991, whose three configurations failed. The selected model remains scientifically provisional.
Model extraction	Full final-2009 numerical fidelity cannot be verified from the preliminary report. Accepted canonical inputs retained. Existing runtime normalizes 40 diet cells (maximum change 0.0003601), uses default GS, derives detritus PB/QB and BA, and includes import 28. Detritus BA is 5934.4585 versus source 5934.798; partially routed whale flows become export. Source/runtime review Transformation ledger
SPPR calculation	[MANUAL — Researcher: calculation choices and reasons] [Excluded groups, scope and relevant values; other adjustments]

Field	Entry
GE diagnostics	No negative SPPR entries across the basal-source columns, including unfished groups. rho_living=0.16615875; b=0.36940829; Detritus (27) SPPR=2.09749784.
TE diagnostics	No negative SPPR entries across the basal-source columns, including unfished groups. rho_living=0.99071698; Detritus (27) SPPR=1.60803192.
Geographic fit	Approximate A: 95–100% of the region covered by the study. B: 20–35% of the study inside the region.
Temporal fit	Static 1987 coefficients and group-catch mixture proxies are applied across 1950–2019. The 2019 reference is 32 years later; publication 2009 does not update ecosystem conditions. Annual ecosystem change is not reconstructed.
Other	All configurations retain WARN and strict balance false. TE is near singular for Coastal tunas (17); ten whale/seabird groups have EE=0. With Egestion has no negative entries, rho_living=0.13292700, b=0.36946627, Detritus SPPR=2.09837460. Source-version, broad offshore-domain and assumed composition limits preclude production approval. Full matrix evidence
Open issues and next action	[Researcher: open issues and next actions]
Review and reproducibility	Researcher name: [name] review date: [date]

Taxon mapping and coverage

Reference: 2019 landings; complete catch-label universe, including unidentified and zero catch; no group filter.

27 source ecological groups (+1 computational import); 512 catch labels.

Simple-chain PPR totals 244,709,206.37 t C in 2019.

100 labels lack a saved classic coefficient/TL; all have zero 2019 landings and therefore zero annual PPR.
Unavailable TL remains ?, not zero.

Method: independent classic/simple trophic chain using the saved taxon coefficient.

[Full taxon mapping appendix](#) and [descriptive Sources sheet](#)

Overall confidence	Taxa	Catch percentage	PPR percentage
High	0	0.0000%	0.0000%
Medium	476	77.4228%	88.8147%
Low	1	0.0002%	0.0002%
Very low	35	22.5770%	11.1850%
Unresolved	0	0.0000%	0.0000%

The summary uses the weaker required membership or allocation confidence. The following tables show each component separately.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Named precursor membership transferred through the verified source-to-model crosswalk; final-source version unverified.	Medium	65.6489%
Complete taxonomic/reporting candidate union; the caught species mixture is inferred.	Medium	11.6131%
Taxonomic extension from actual precursor species, genera or family representatives.	Medium	11.5528%
Broad reporting category approximated over compatible source guilds; composition unknown.	Very low	10.9590%
Positive taxonomic or habitat analogue with an explicit source-pool mismatch.	Very low	0.2260%
Actual bony members support the placement, but a contradictory source family label limits certainty.	Low	0.0002%

Membership evidence: [precursor members, taxonomy and candidate decisions](#) Bony reporting excludes dedicated Sharks/Rays, while genuine mixed guilds remain eligible. Elasmobranchii excludes chimaeras; sawfish membership in groups 14/15 is retained. Scaridae denotes the historical parrotfish family, now Scarinae within Labridae; the genus-misspelling homonym is excluded. [Scaridae rank bridge](#)

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
One reviewed group; no numerical split.	High	71.9319%
Complete accepted 1987 group-export catch proportions; fixed historical composition assumed.	Medium	28.0681%

Each rule percentage uses the complete independent simple-chain PPR denominator, once per taxon. Weights and assumptions: all 49 multi-group labels use complete accepted 1987 model-export catches at Medium confidence, fixed across 1950–2019 and all catch bases; native fleet data and modern mixtures are unavailable.

[Complete allocation evidence](#)

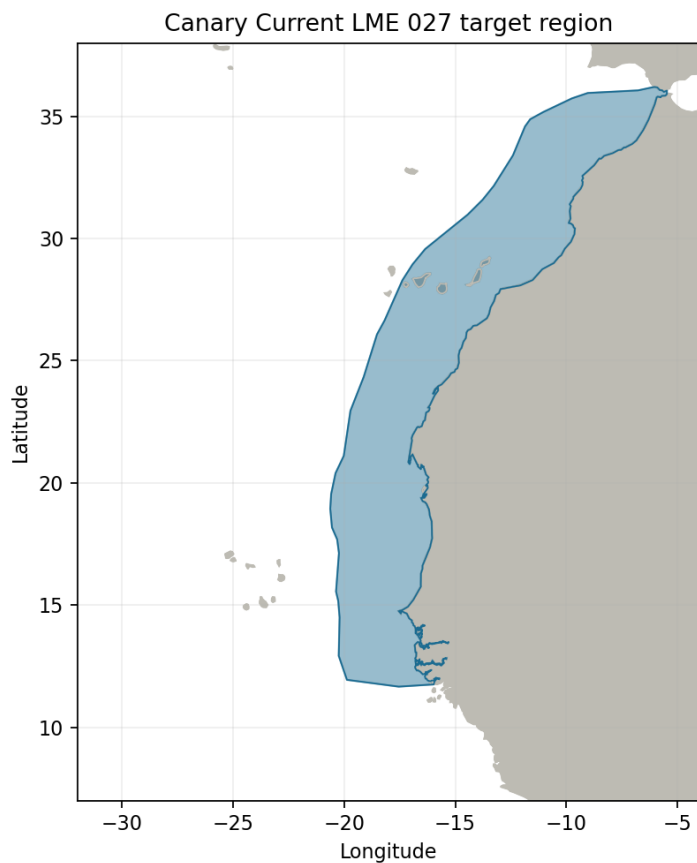
No unresolved taxa remain. Very low: 35 labels, 22.5770% of landings and 11.1850% of classic PPR. Broad fish/invertebrate mixtures and unlisted habitat/taxonomic analogues retain explicit assumptions in the appendix. Centrolophidae is Low: named Schedophilus members support groups 12/18, while its contradictory Sharks-15 label remains unresolved. Separately, Marine fishes not identified contributes 20.47% of GE, 40.70% of TE and 20.58% of With Egestion model PPR. TE totals 1.946 billion t C; its extreme source coefficients are a diagnostic concern, not evidence of reliable food-web demand. [Method-specific contribution evidence](#)

Geographic evidence screenshots

Target R is the shelf LME; study S is the broader offshore ecosystem. Bounds-based marine reconstructions bracket source-version ambiguity; these are coarse geographic approximations, not exact author polygons or statistical intervals.

Region area

Northwest African Shelf target region

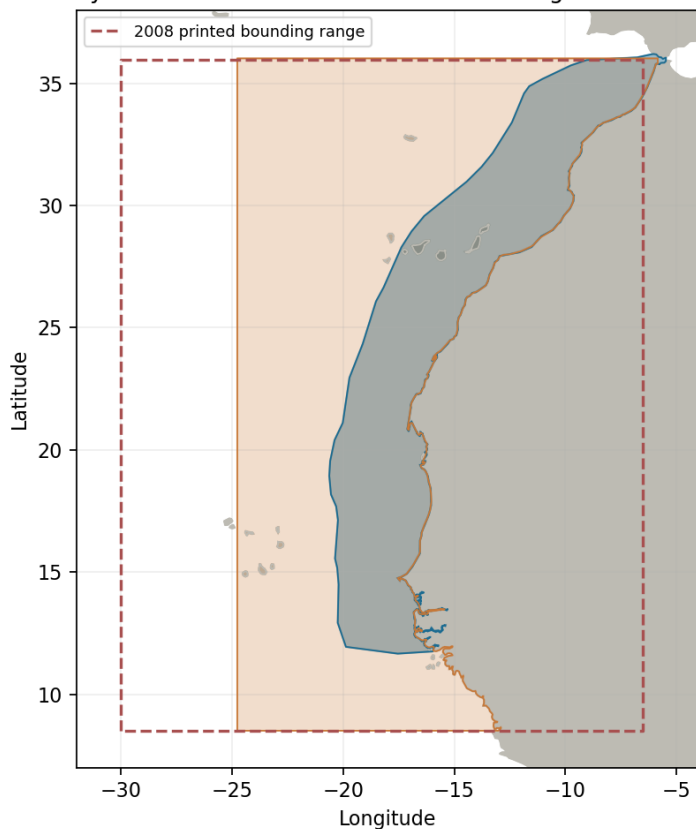


Sea Around Us LME 27 target R: 1,123,612 km². [Target boundary](#) | [Map image](#)

Study area in the article

Inherited and preliminary source-bound reconstructions

Canary Current LME 027 and source bounding reconstructions



EcoBase 118 extent and 2008 preliminary bounds differ in longitude. Marine reconstructed domains are about 3.43 and 5.08 million km², with A about 99% and B about 22–33%. The reported model area is a separate 3.56 million km² quantity. The broad 95–100% / 20–35% ranges reflect boundary/version uncertainty. [Geographic assessment](#) | [Portable reconstruction](#)

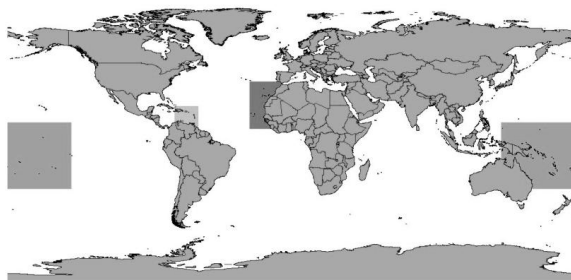
Original 2008 preliminary Figure 1, PDF 29 (caption PDF 28), rather than the unavailable final-2009 figure. Panel A shows global project areas; panel B shows Northwest African bounds of 8.5–36°N, 30–6.5°W. [Preliminary source figure](#)

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1 **Figure 1**

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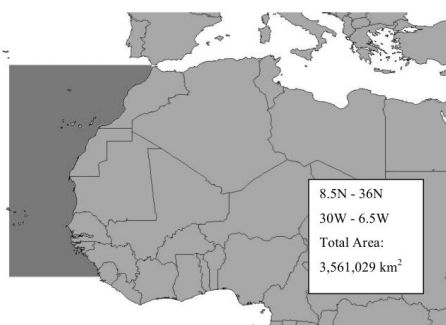
A)



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4

B)



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